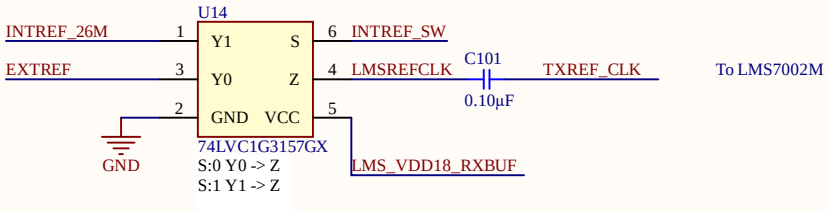
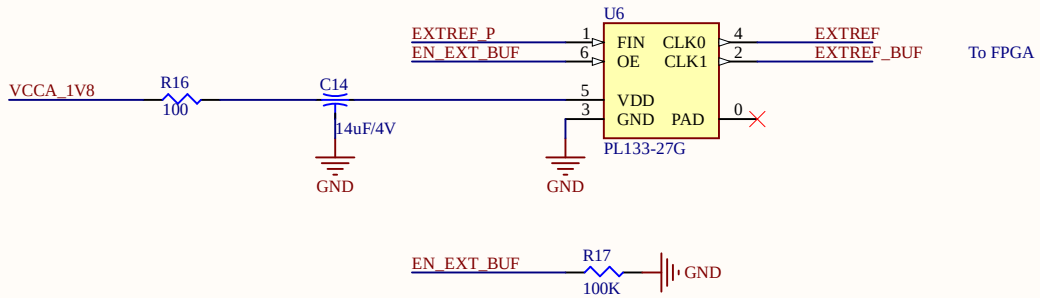
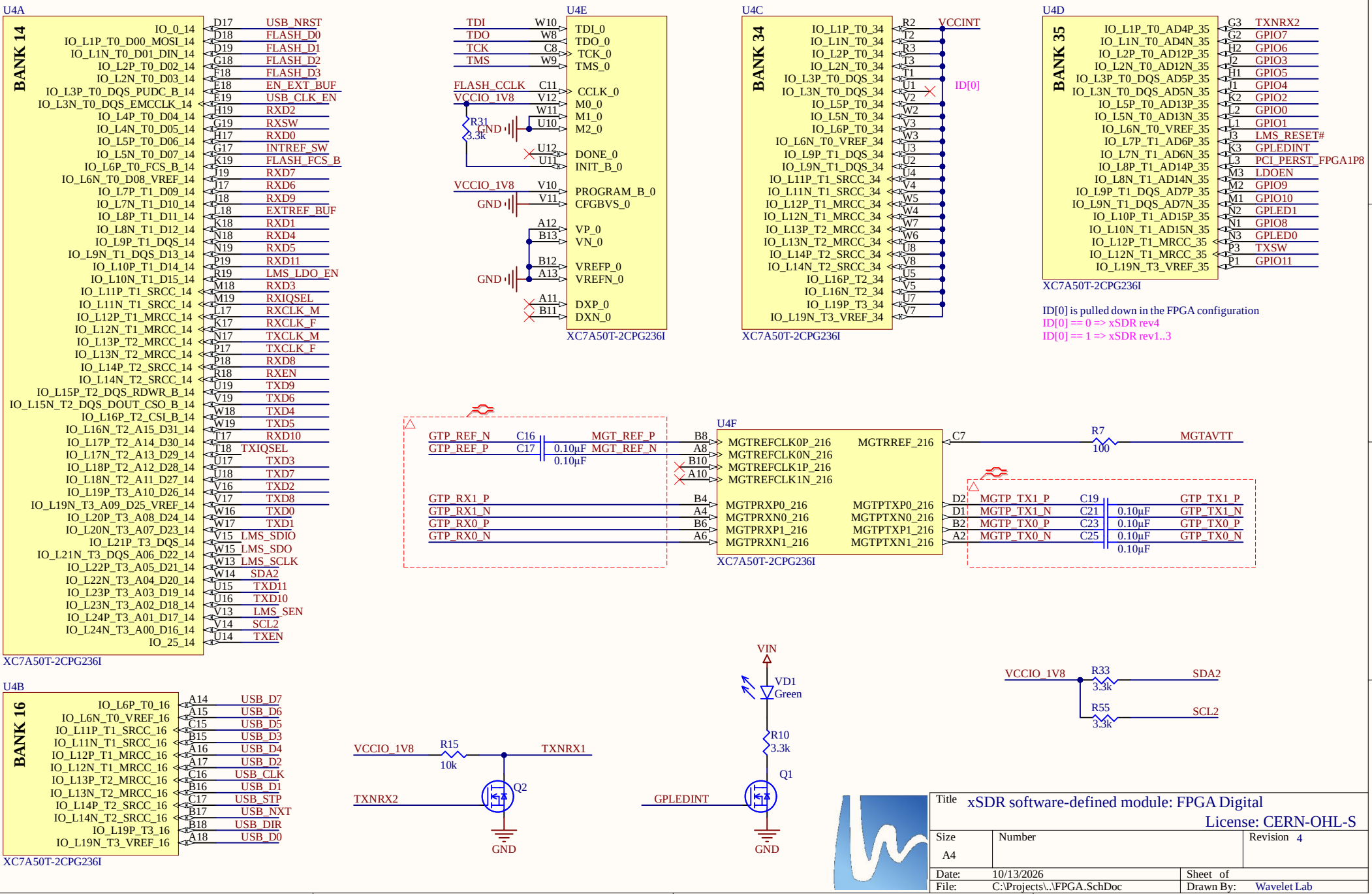


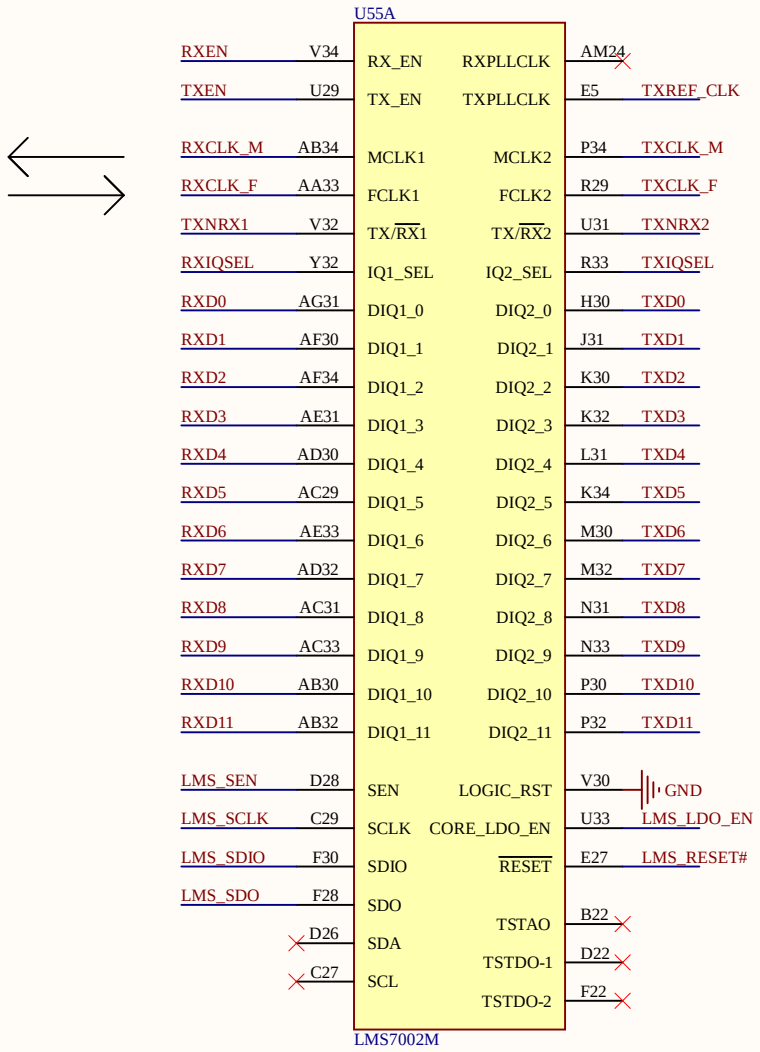
This is an analog switch, not a buffer!



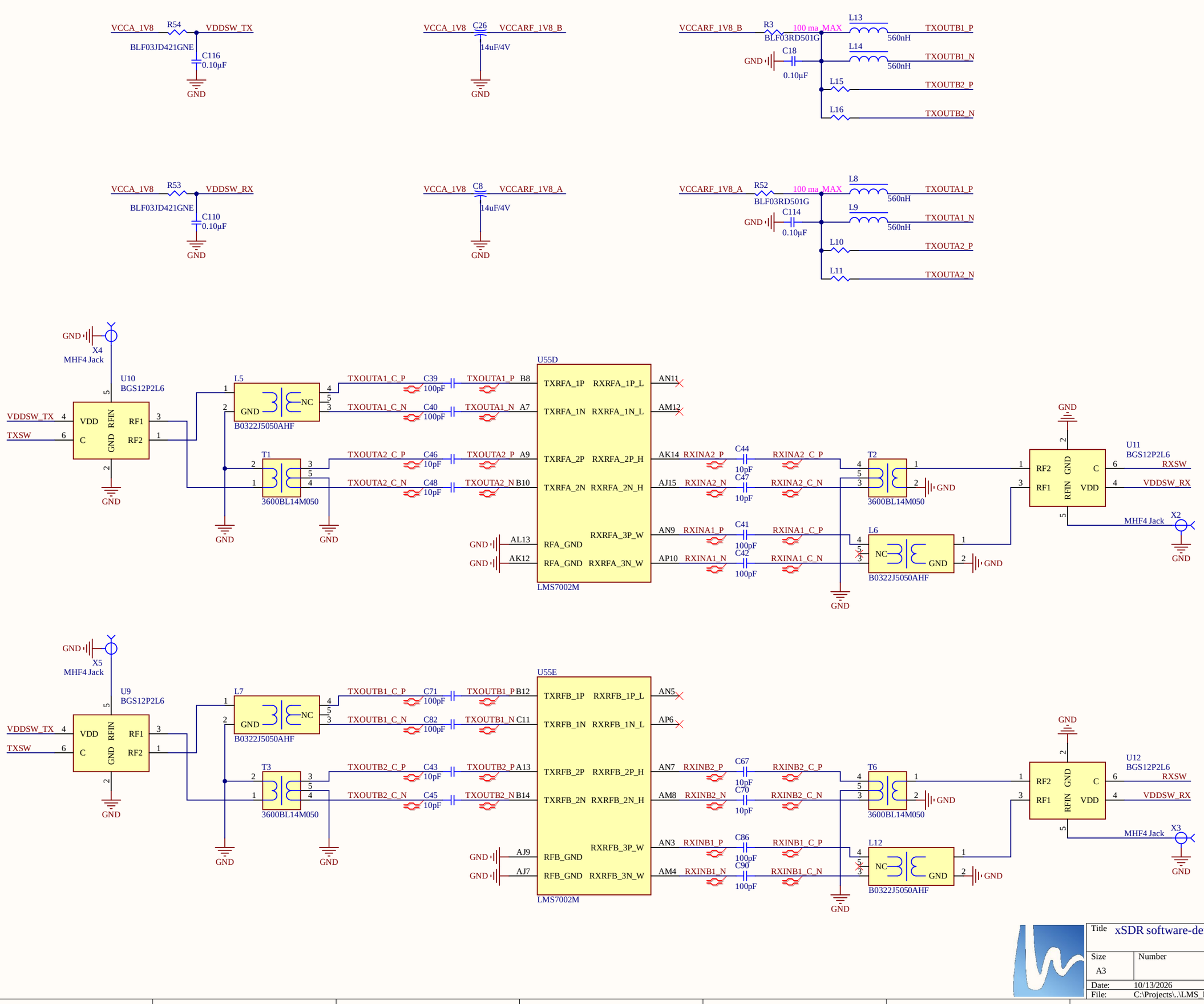


Title	xSDR software-defined module: clocking License: CERN-OHL-S	
Size: A4	Number	Revision: 4
Date: 10/13/2026		Sheet of
File: C:\Projects\..\Clocking.SchDoc		Drawn By: Wavelet Lab

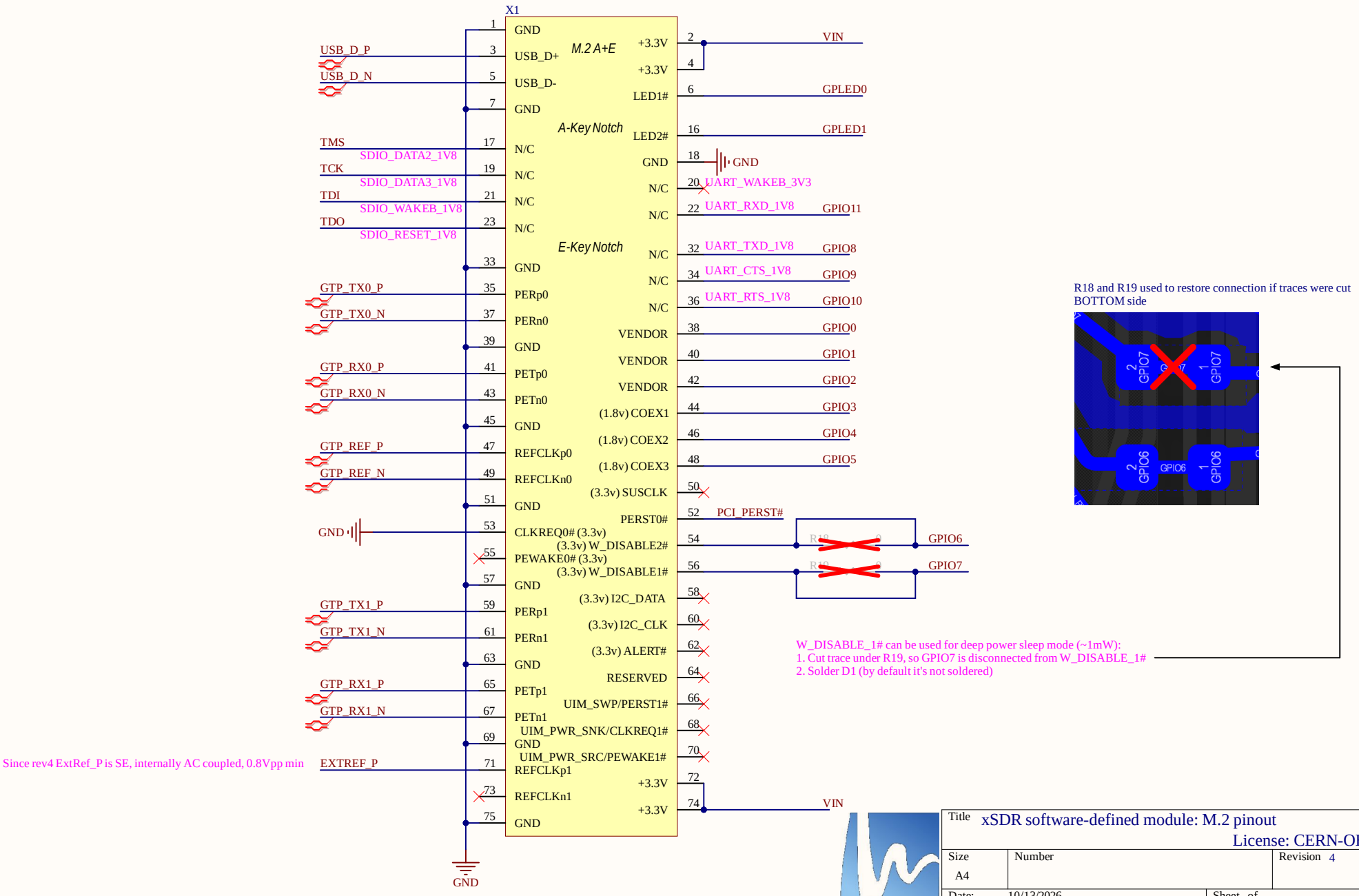

U4A
BANK 14
IO_0_14 D17 USB_NRST
IO_L1P_T0_D00_MOSI_14 D18 FLASH_D0
IO_L1N_T0_D01_DIN_14 D19 FLASH_D1
IO_L2P_T0_D02_14 G18 FLASH_D2
IO_L2N_T0_D03_14 F18 FLASH_D3
IO_L3P_T0_DQS_PUDC_B_14 E18 EN_EXT_BUF
IO_L3N_T0_DQS_EMCCLK_14 E19 USB_CLK_EN
IO_L4P_T0_D04_14 H19 RXD2
IO_L4N_T0_D05_14 G19 RXSW
IO_L5P_T0_D06_14 H17 RXD0
IO_L5N_T0_D07_14 G17 INTREF_SW
IO_L6P_T0_FCS_B_14 K19 FLASH_FCS_B
IO_L6N_T0_D08_VREF_14 J19 RXD7
IO_L7P_T1_D09_14 J17 RXD6
IO_L7N_T1_D10_14 J18 RXD9
IO_L8P_T1_D11_14 L18 EXTREF_BUF
IO_L8N_T1_D12_14 K18 RXD1
IO_L9P_T1_DQS_14 N18 RXD4
IO_L9N_T1_DQS_D13_14 N19 RXD5
IO_L10P_T1_D14_14 P19 RXD11
IO_L10N_T1_D15_14 R19 LMS_LDO_EN
IO_L11P_T1_SRCC_14 M18 RXD3
IO_L11N_T1_SRCC_14 M19 RXIQSEL
IO_L12P_T1_MRCC_14 L17 RXCLK_M
IO_L12N_T1_MRCC_14 K17 RXCLK_F
IO_L13P_T2_MRCC_14 N17 TXCLK_M
IO_L13N_T2_MRCC_14 P17 TXCLK_F
IO_L14P_T2_SRCC_14 P18 RXD8
IO_L14N_T2_SRCC_14 R18 RXEN
IO_L15P_T2_DQS_RDWR_B_14 U19 TXD9
IO_L15N_T2_DQS_DOUT_CSO_B_14 V19 TXD6
IO_L16P_T2_CSI_B_14 W18 TXD4
IO_L16N_T2_A15_D31_14 W19 TXD5
IO_L17P_T2_A14_D30_14 T17 RXD10
IO_L17N_T2_A13_D29_14 T18 TXIQSEL
IO_L18P_T2_A12_D28_14 U17 TXD3
IO_L18N_T2_A11_D27_14 U18 TXD7
IO_L19P_T3_A10_D26_14 V16 TXD2
IO_L19N_T3_A09_D25_VREF_14 V17 TXD8
IO_L20P_T3_A08_D24_14 W16 TXD0
IO_L20N_T3_A07_D23_14 W17 TXD1
IO_L21P_T3_DQS_14 V15 LMS_SDIO
IO_L21N_T3_DQS_A06_D22_14 W15 LMS_SDO
IO_L22P_T3_A05_D21_14 W13 LMS_SCLK
IO_L22N_T3_A04_D20_14 W14 SDA2
IO_L23P_T3_A03_D19_14 U15 TXD11
IO_L23N_T3_A02_D18_14 U16 TXD10
IO_L24P_T3_A01_D17_14 V13 LMS_SEN
IO_L24N_T3_A00_D16_14 V14 SCL2
IO_25_14 U14 TXEN
XC7A50T-2CPG236I
U4B
BANK 16
IO_L6P_T0_16 A14 USB_D7
IO_L6N_T0_VREF_16 A15 USB_D6
IO_L11P_T1_SRCC_16 C15 USB_D5
IO_L11N_T1_SRCC_16 B15 USB_D3
IO_L12P_T1_MRCC_16 A16 USB_D4
IO_L12N_T1_MRCC_16 A17 USB_D2
IO_L13P_T2_MRCC_16 C16 USB_CLK
IO_L13N_T2_MRCC_16 B16 USB_D1
IO_L14P_T2_SRCC_16 C17 USB_STP
IO_L14N_T2_SRCC_16 B17 USB_NXT
IO_L19P_T3_16 B18 USB_DIR
IO_L19N_T3_VREF_16 A18 USB_D0
XC7A50T-2CPG236I
U4E
TDI W10 TDI_0
TDO W8 TDO_0
TCK C8 TCK_0
TMS W9 TMS_0
FLASH_CCLK C11 CCLK_0
VCCIO_1V8 V12 M0_0
W11 M1_0
R31 3.3k GND U10 M2_0
U12 DONE_0
U11 INIT_B_0
VCCIO_1V8 V10 PROGRAM_B_0
GND V11 CFGBVS_0
A12 VP_0
B13 VN_0
B12 VREFP_0
GND A13 VREFN_0
A11 DXP_0
B11 DXN_0
XC7A50T-2CPG236I
U4C
BANK 34
IO_L1P_T0_34 R2 VCCINT
IO_L1N_T0_34 T2
IO_L2P_T0_34 R3
IO_L2N_T0_34 T3
IO_L3P_T0_DQS_34 T1
IO_L3N_T0_DQS_34 U1 ID[0]
IO_L5P_T0_34 V2
IO_L5N_T0_34 W2
IO_L6P_T0_34 V3
IO_L6N_T0_VREF_34 W3
IO_L9P_T1_DQS_34 U3
IO_L9N_T1_DQS_34 U2
IO_L11P_T1_SRCC_34 U4
IO_L11N_T1_SRCC_34 V4
IO_L12P_T1_MRCC_34 W5
IO_L12N_T1_MRCC_34 W4
IO_L13P_T2_MRCC_34 W7
IO_L13N_T2_MRCC_34 W6
IO_L14P_T2_SRCC_34 U8
IO_L14N_T2_SRCC_34 V8
IO_L16P_T2_34 U5
IO_L16N_T2_34 V5
IO_L19P_T3_34 U7
IO_L19N_T3_VREF_34 V7
XC7A50T-2CPG236I
U4D
BANK 35
IO_L1P_T0_AD4P_35 G3 TXNRX2
IO_L1N_T0_AD4N_35 G2 GPIO7
IO_L2P_T0_AD12P_35 H2 GPIO6
IO_L2N_T0_AD12N_35 J2 GPIO3
IO_L3P_T0_DQS_AD5P_35 H1 GPIO5
IO_L3N_T0_DQS_AD5N_35 J1 GPIO4
IO_L5P_T0_AD13P_35 K2 GPIO2
IO_L5N_T0_AD13N_35 L2 GPIO0
IO_L6N_T0_VREF_35 L1 GPIO1
IO_L7P_T1_AD6P_35 J3 LMS_RESET#
IO_L7N_T1_AD6N_35 K3 GPLEDINT
IO_L8P_T1_AD14P_35 L3 PCI_PERST_FPGA1P8
IO_L8N_T1_AD14N_35 M3 LDOEN
IO_L9P_T1_DQS_AD7P_35 M2 GPIO9
IO_L9N_T1_DQS_AD7N_35 M1 GPIO10
IO_L10P_T1_AD15P_35 N2 GPLED1
IO_L10N_T1_AD15N_35 N1 GPIO8
IO_L12P_T1_MRCC_35 N3 GPLED0
IO_L12N_T1_MRCC_35 P3 TXSW
IO_L19N_T3_VREF_35 P1 GPIO11
XC7A50T-2CPG236I
ID[0] is pulled down in the FPGA configuration
ID[0] == 0 => xSDR rev4
ID[0] == 1 => xSDR rev1..3
U4F
GTP_REF_N C16
GTP_REF_P C17
0.10µF
0.10µF
MGT_REF_P
MGT_REF_N
B8
A8
B10
A10
MGTREFCLK0P_216
MGTREFCLK0N_216
MGTREFCLK1P_216
MGTREFCLK1N_216
MGTRREF_216
C7
R7
100
MGTAVTT
GTP_RX1_P B4
GTP_RX1_N A4
GTP_RX0_P B6
GTP_RX0_N A6
MGTPRXP0_216
MGTPRXN0_216
MGTPRXP1_216
MGTPRXN1_216
MGTPTXP0_216
MGTPTXN0_216
MGTPTXP1_216
MGTPTXN1_216
D2 MGTP_TX1_P C19 GTP_TX1_P
D1 MGTP_TX1_N C21 0.10µF GTP_TX1_N
B2 MGTP_TX0_P C23 0.10µF GTP_TX0_P
A2 MGTP_TX0_N C25 0.10µF GTP_TX0_N
0.10µF
XC7A50T-2CPG236I
VCCIO_1V8 R15 10k TXNRX1
TXNRX2 Q2
GND
VIN
VD1 Green
R10 3.3k
Q1
GPLEDINT
GND
VCCIO_1V8 R33 3.3k SDA2
R55 3.3k SCL2
Title xSDR software-defined module: FPGA Digital
License: CERN-OHL-S
Size A4
Number
Revision 4
Date: 10/13/2026
Sheet of
File: C:\Projects\..\FPGA.SchDoc
Drawn By: Wavelet Lab



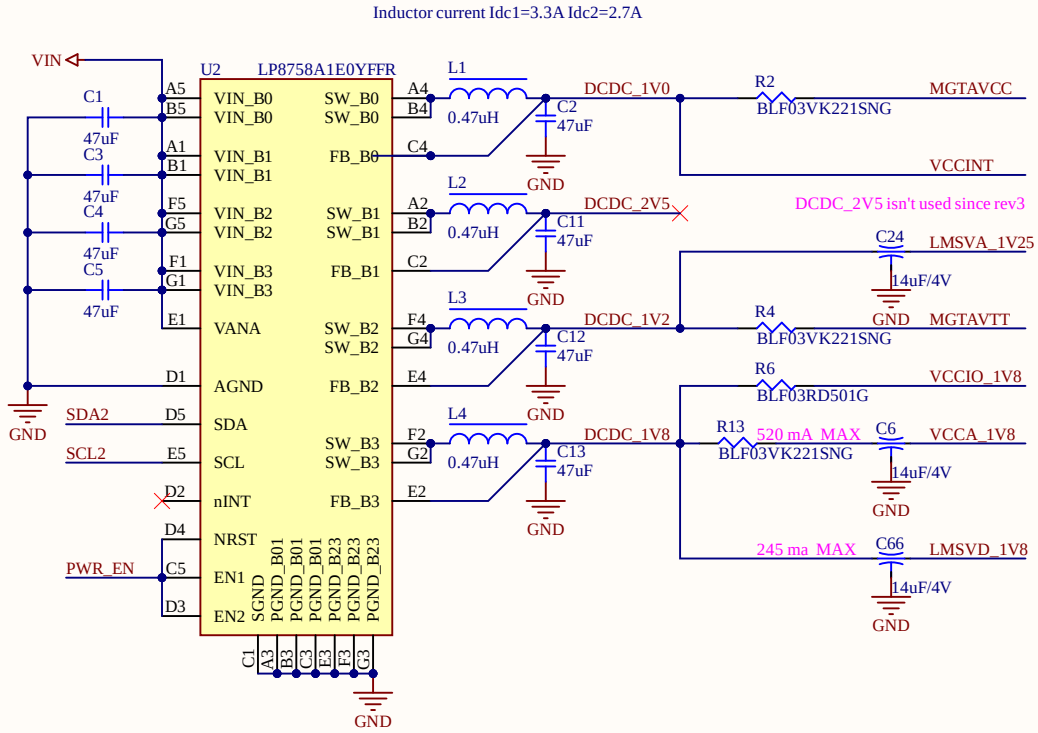
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File: C:\Projects\..\LMS_Dig.SchDoc		Drawn By: Wavelet Lab



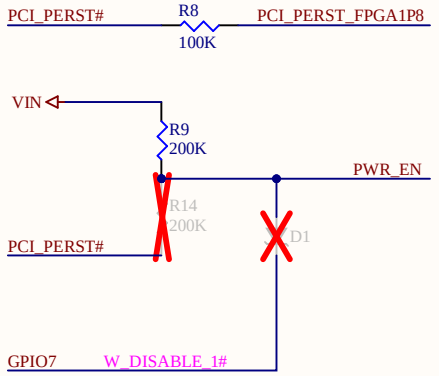
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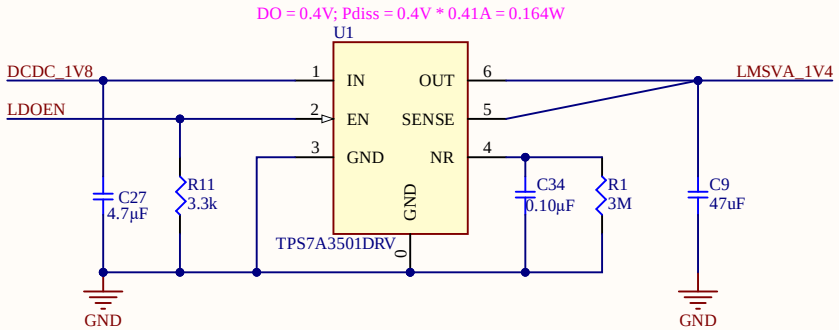


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Date: 10/13/2026		Sheet of
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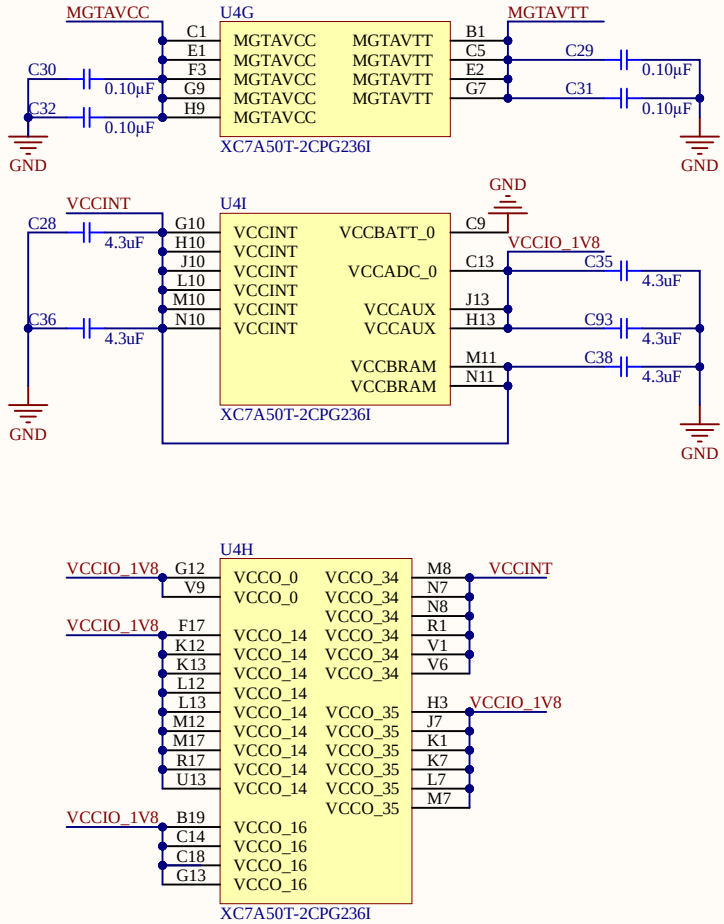


BLF03JD421GNE -- 370mA MAX
BLF03RD501G -- 380mA MAX
BLF03VK221SNG -- 500mA MAX
BLM03PG330SN1 -- 750mA MAX
BLM03HD182FN1 -- 100mA MAX

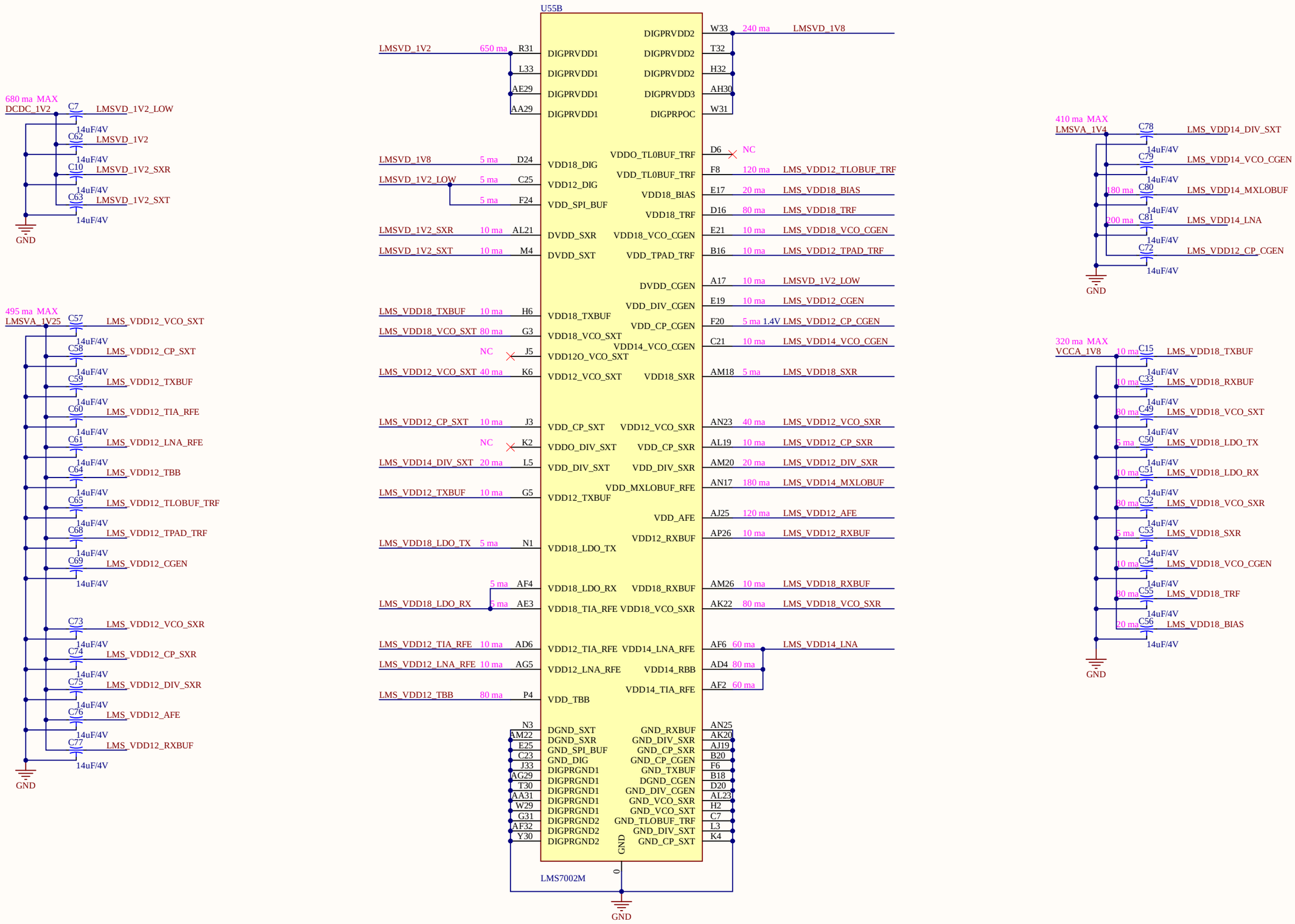




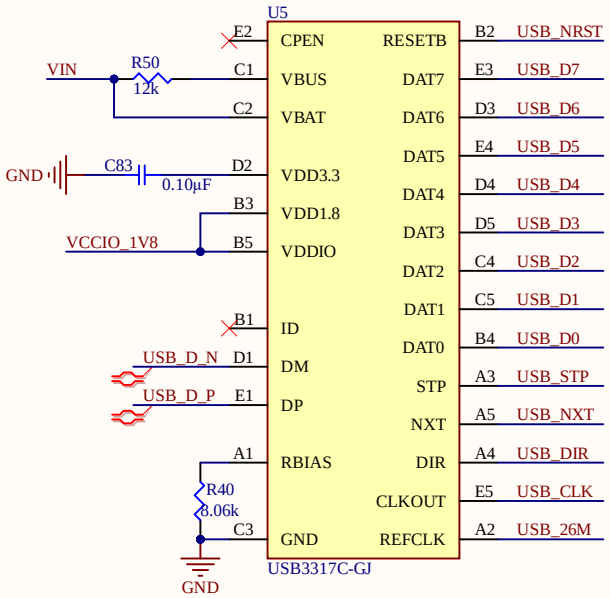
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Size: A4	Number	Revision: 4
Date: 10/13/2026		Sheet of
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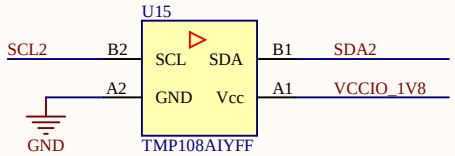


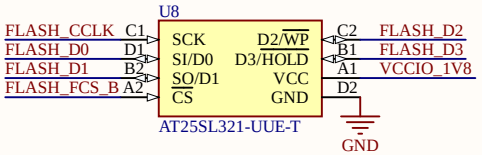
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Date: 10/13/2026		Sheet of
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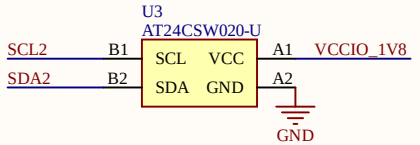
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	License: CERN-OHL-S		
Size	Number		Revision 4
A3			
Date:	10/13/2026	Sheet of	
File:	C:\Projects\..\PwrLMS7002M.SchDoc	Drawn By:	Wavelet Lab











Title	xSDR software-defined module: USB & AUX logic License: CERN-OHL-S	
Size: A4	Number	Revision: 4
Date: 10/13/2026		Sheet of
File: C:\Projects\..\USB2_Aux.SchDoc		Drawn By: Wavelet Lab